

CS471 – Study Guide

Lessons 32-37: CNN Applications, LLMs, RAG, Ethics, & Agentic AI

Study Strategy

This study guide synthesizes the final block of the course, moving from advanced computer vision applications to the cutting edge of Generative AI and the ethical vulnerabilities of these systems. Focus heavily on architectural leaps (Self-Attention, Vector Embeddings, ReAct Loops) and the critical failure modes that commanders must mitigate.

Lesson 32: CNN Applications

Beyond basic image classification, CNN architectures are adapted for complex tactical battlespace analysis.

- **Object Detection (e.g., YOLO):** Not only classifies objects but draws a *bounding box* around them, outputting exact spatial coordinates. Crucial for counting multiple targets in a single drone feed.
- **Semantic Segmentation:** Assigns a class to *every single pixel* in the image. Used for highly precise boundary mapping (e.g., highlighting exactly where the road ends and the mud begins).
- **Transfer Learning:** Training a deep CNN from scratch requires massive data and compute. Transfer learning takes a model pre-trained on a massive civilian dataset, freezes its foundational feature-extraction layers, and only retrain the final classification layers on a small, classified military dataset.

Lesson 33: Transformers and LLMs

The Architectural Leap

- **Recurrent Neural Networks (RNNs):** Older models that read text linearly (left-to-right). They suffered from severe memory loss over long paragraphs.
- **The Transformer & Self-Attention:** Processes all words in a sequence *simultaneously*. The Self-Attention mechanism mathematically calculates how strongly every word relates to every other word, perfectly capturing complex context.

Next-Token Prediction

An autoregressive Large Language Model (LLM) is a **statistical prediction engine**. It calculates a probability distribution across its vocabulary and samples the most mathematically likely *next token* (word/sub-word), appends it, and loops until finished.

- **Hallucination:** Because the model is predicting statistically likely text rather than retrieving verified facts, it will occasionally generate highly plausible, grammatically perfect, but completely fabricated information.

Lesson 34: Prompt Engineering

- **Zero-Shot vs. Few-Shot Prompting:** Zero-shot asks the model to perform a task with zero prior examples. Few-shot provides a few explicit input-output examples in the prompt to teach it the desired format or logic.
- **Chain-of-Thought (CoT):** Forcing the model to generate intermediate reasoning steps (e.g., "Think step-by-step"). This gives the model more generated tokens to process its internal logic before committing to a complex answer.
- **Prompt Injection:** An adversarial attack where a user inputs a prompt that overrides the LLM's core system instructions. *Indirect Prompt Injection* occurs when an LLM reads an external webpage containing hidden malicious text commands.

Lesson 35: Retrieval-Augmented Generation (RAG)

Fine-tuning an LLM on new classified data is incredibly expensive and embeds the data opaquely in the weights. **RAG** solves this by giving the LLM the ability to "read" external documents dynamically.

- **Vector Embeddings:** Text is converted into high-dimensional mathematical vectors. Words or sentences with similar semantic meanings sit close to each other in this mathematical space.
- **The RAG Pipeline:**
 1. User asks a question.
 2. The system converts the question into a vector and performs a *similarity search* in a **Vector Database** containing military manuals.
 3. The system retrieves the most relevant text chunks.
 4. Those chunks are injected directly into the LLM's context window, forcing the model to ground its answer in the provided citations and preventing hallucinations.

Lesson 36: Failure Modes and Ethics

Commanders must understand that AI systems are brittle and vulnerable to both intentional attacks and systemic structural failures.

- **Algorithmic Bias:** AI models are trained on historical human data. If that data contains societal prejudices or unequal representation, the model will statically learn, reflect, and mathematically amplify those biases in its tactical outputs.
- **The Alignment Problem:** The extreme difficulty of ensuring an AI's mathematical objective function perfectly matches the human operator's true intent. An unaligned AI might find dangerous, unintended loopholes to maximize its reward score at the expense of safety.

- **Data Poisoning:** An adversarial attack during the training phase. The adversary subtly alters the training dataset (e.g., changing a few pixels on images of enemy tanks) so the model learns a hidden "backdoor" and misclassifies targets in combat.
- **The Dual-Use Dilemma:** The ethical friction inherent in AI research, where foundational models built for positive civilian applications (like medical diagnosis or search-and-rescue navigation) can be easily repurposed for lethal autonomous weapons.

Lesson 37: Agentic AI

Standard LLMs are passive text generators. **Agentic AI** elevates the model by giving it autonomy and the ability to use external **Tools** (e.g., Python interpreters, web browsers, databases).

The ReAct (Reason + Act) Loop

Instead of immediately outputting an answer, an Agent enters a recursive operational loop:

1. **Reason:** The agent synthesizes its internal logic to decide what information it lacks or what tool it needs to use next.
2. **Act:** The agent physically executes a tool (e.g., writing and running a Python script).
3. **Observe:** The agent parses the exact returned output from the tool.

The Agent repeats this Reason-Act-Observe loop until it determines it has acquired enough verified evidence to synthesize a final, accurate response.

Key Concepts Reference

1. **Self-Attention:** Weighs the contextual importance of all words in a sequence simultaneously.
2. **Next-Token Prediction:** The autoregressive statistical generation of text.
3. **RAG:** Grounding LLM responses in dynamically retrieved vector-database documents.
4. **Algorithmic Bias:** The amplification of historical prejudices found in training data.
5. **Agentic AI:** LLMs using the ReAct loop to autonomously execute tools and solve problems.